COSC3106: Theory of Computing

Lec#1: Introduction

School of Computer Science and Technology

Algoma UNIVERSITY

Purpose and motivation

- This course is on the "Theory of Computation". It is all about the mathematical properties of computer hardware and software, which tries to answer the following questions:

1) What is an algorithm?
   - The algorithm is a finite sequence of elementary steps.
   - By the end of this course, you will have a formal definition of what an algorithm is.

2) Can "everything" be solved by an algorithm (the limitations of computers)?
   - We will see later, the answer to this question is "no".

3) Can all math problems be solved in a systematic way?
   - "in a systematic way" means by an algorithm
   - In other words, the question was "does there exist one systematic way you can use to solve all possible problems in math so all problems can be formulated.

- **The theory of computation consists of three areas:**
  1) **Complexity Theory**
     - Why some problems easy?
       - "**Easy**": If it is efficiently solvable, which means that there exists fast algorithm to solve the problem.
     - Why some problems hard?
       - "**Hard**": If it cannot be solved efficiently, or if we don't know whether it can be solved efficiently, means there no exists fast algorithm to solve the problem?
  - Examples:
    - Easy:
      a) Sorting a sequence of, say, 1,000,000 numbers,
      b) Searching for a name in a telephone directory, and,
    - Hard:
      a) Timetable scheduling for all courses at a university,
      b) Factoring a 300-digit integer into its prime factors.

- **Central Question in Complexity Theory:**
  - Classify problems according to their degree of “difficulty”. Give a rigorous proof that problems that seem to be “hard” are really “hard”.

2) Computability Theory

- Is an arbitrary mathematical statement true or false?

- Examples:
  - Is there a student in this classroom whose ID number is 123456? (solvable problem)
  - Does a given computer program run forever or eventually stop? (unsolvable problem)

- **Central Question in Computability Theory:**
  - Classify problems as being solvable or unsolvable.
  - The theoretical models that were proposed in order to understand solvable and unsolvable problems led to the development of real computers.

3) Automata Theory

- Automata Theory deals with definitions and properties of different types of "computation models".

- Examples:
  - Finite Automata.
  - Context-Free Grammars.
  - Turing Machines.

- **Central Question in Automata Theory:**
  - Do these models have the same power, or can one model solve more problems than the other?

- **In this course, we will study the last two areas in reverse order:**
  - We will start with:
    - Automata Theory.
  - Followed by:
    - Computability Theory.
  - However, the first area,
    - Complexity Theory, will be covered in Algorithms.

How do we support your learning?

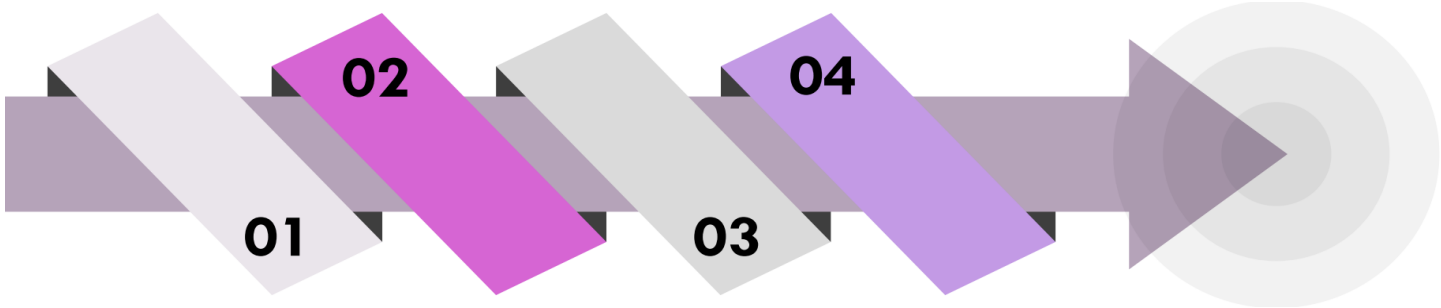
- Instructor: Dr. Omar Al-Buraiki
- Email: omar.alburaiki@algomau.ca
- Office 24 Queen St, #202A; Office hours: Friday from 3-4pm, or online, (email me for an appointment)
- Lectures: Friday: 12:00-3:00pm @ 305E-56 Queen St.

Course description:

- Introduction and Distribution of Outline
- Introduction to Finite Automata
- Deterministic and Non-Deterministic Finite Automata
- Regular Expression (RE) and Regular Language.
- Closure Properties of RE
- Introduction to Context Free Grammars and Languages
- Push Down Automata
- Chomsky Normal Form and the Pumping Lemma for CFL's
- Turing Machines
- Classical Undecidability Problems
- Closure Properties and Post Correspondence.

Intended Learning Outcomes:

- At the end of the course, you are aimed to be able to:



01 Understand the fundamental mathematical basis for today's computing systems.

02 Understand the various languages processed by the application of the finite state machines.

03 Understand how to represent languages.

04 Review classic computability problems.

- **Course Information on BrightSpace:**
  - https://remote.algomau.ca/d2l/home/
  - Regularly check the course website on BrightSpace for announcements.

- The **primary resource** for this course:
  - Required Course Text: John E. Hopcroft, Rajeev Motwani and Jeffery D. Ullman, "Introduction to Automata Theory, Languages, and Computation", 2nd Edition, Addison Wesley/Pearson.

- The provided course slides, class notes, and the textbook are the primary resource for learning this course.

- **Grading Scheme:**
  - Midterm Exam (30%)
  - Final Exam (50%)
  - Quizzes (20%)

- **Exams:**
  - Missing the midterm or a quiz will result in receiving a mark of 0, unless there is a valid documented reason.
  - If a valid reason is provided for absence, a proper accommodation will be considered.